

ExpenseRatio



5.00%



Commission Rate



5.00%



Insurance Analytics Portfolio Project

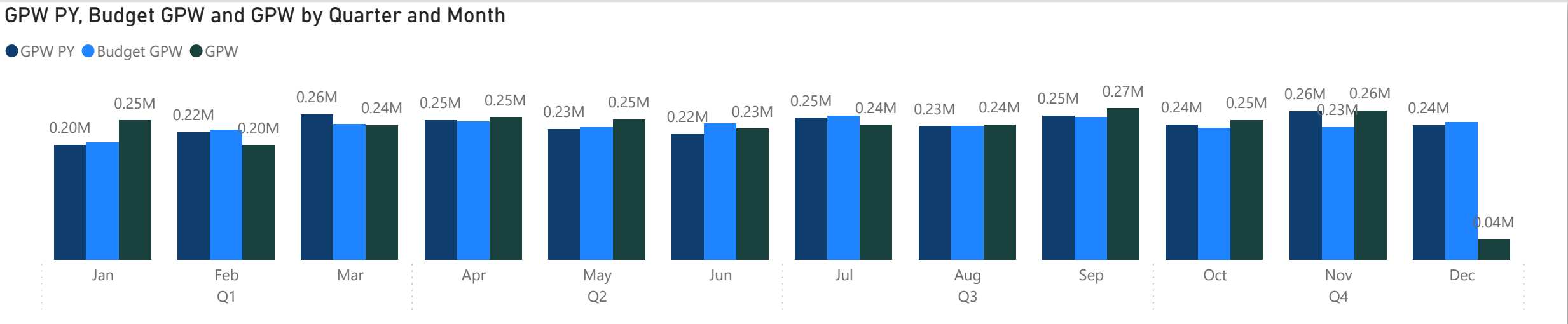
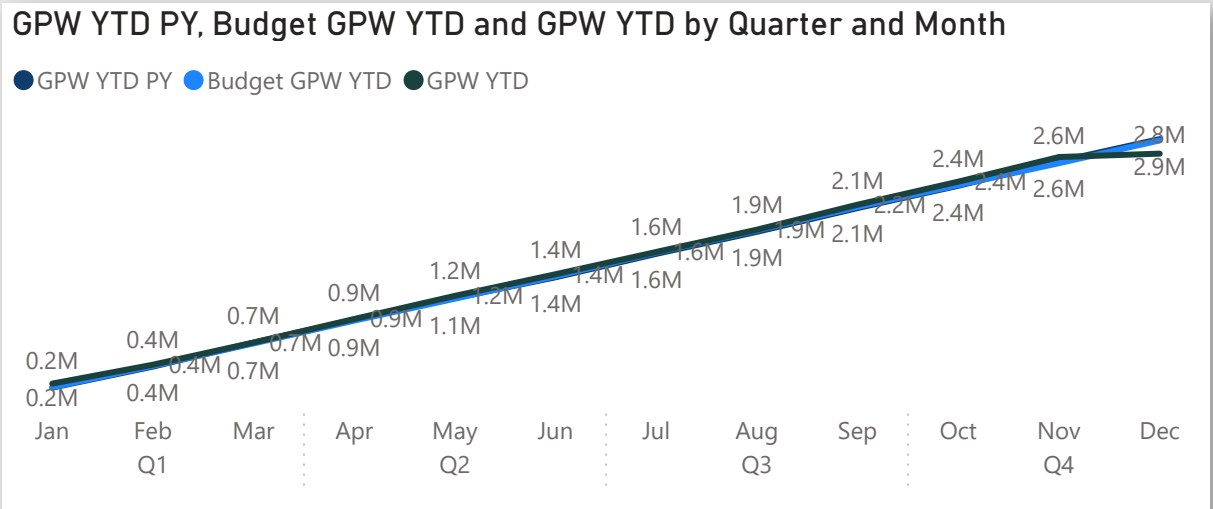
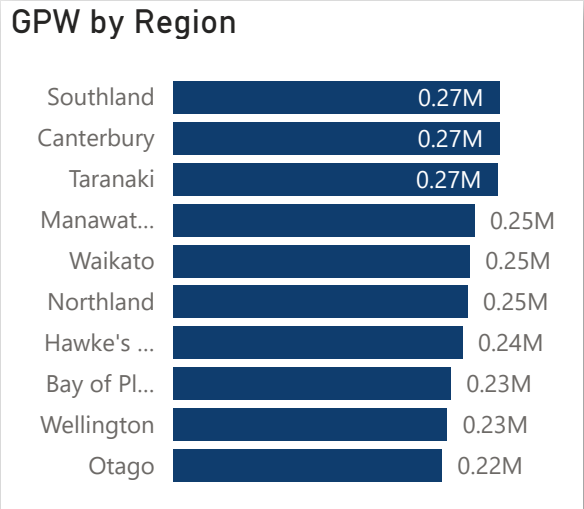
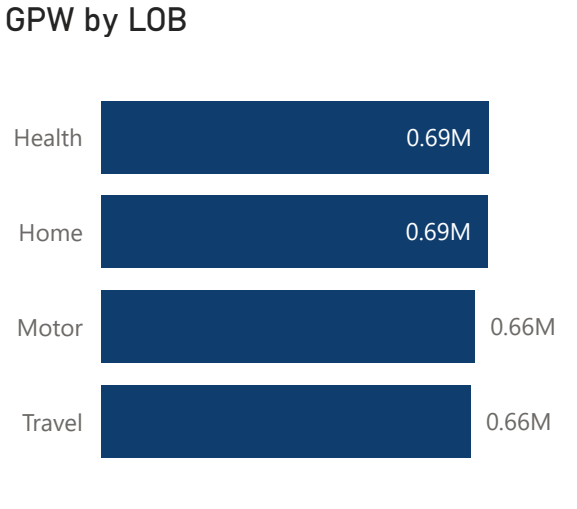
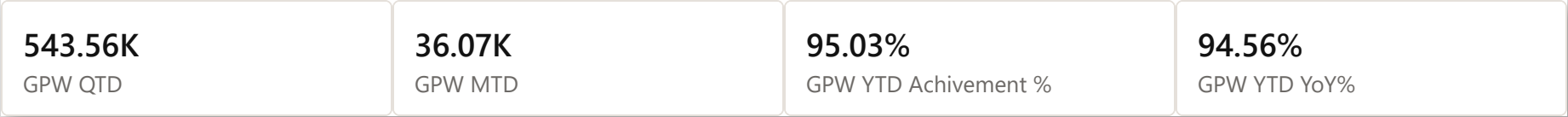
This Power BI dashboard demonstrates a comprehensive analysis of key insurance metrics, covering **premiums, losses, claims, customers, lines of business (LOBs), products, and regional performance**.

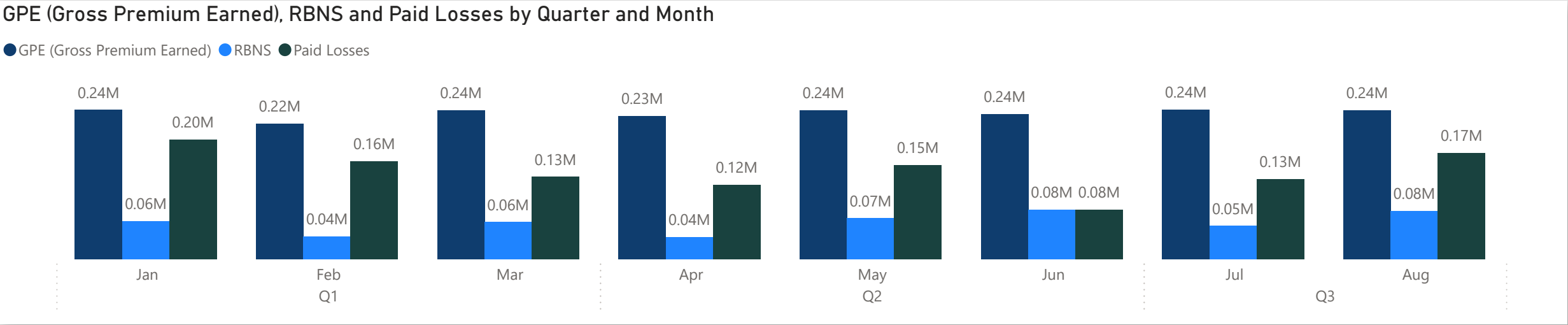
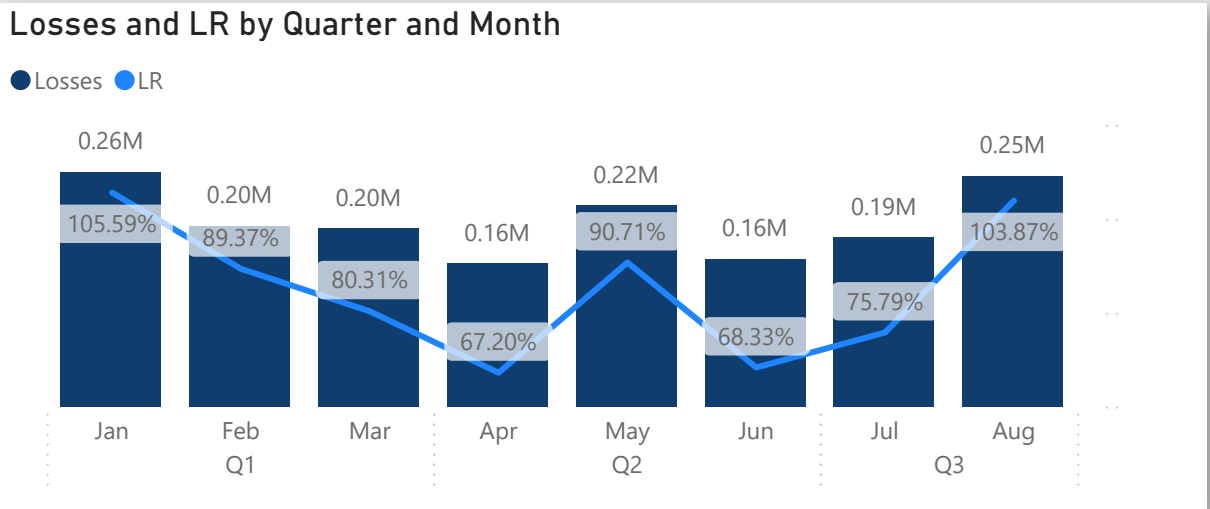
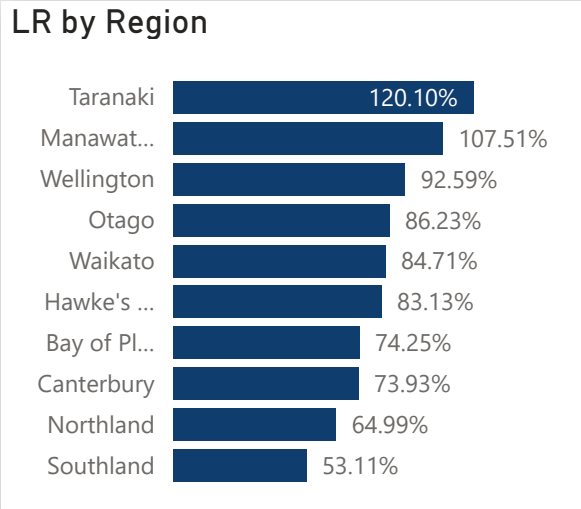
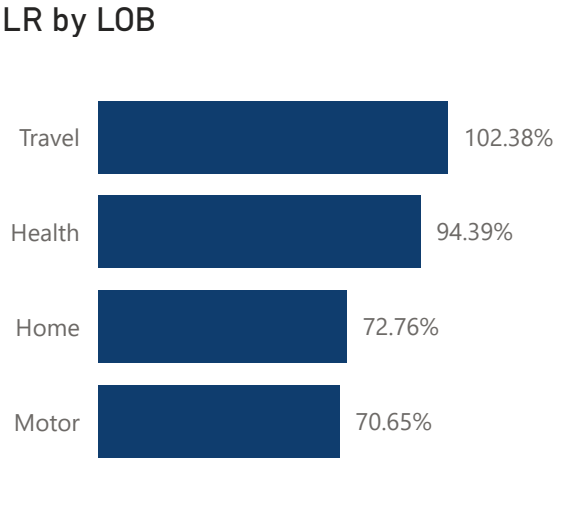
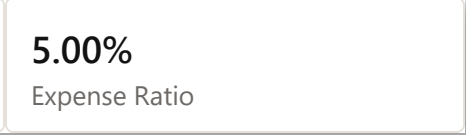
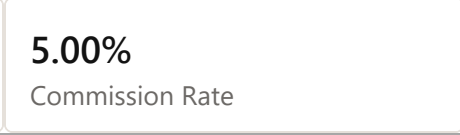
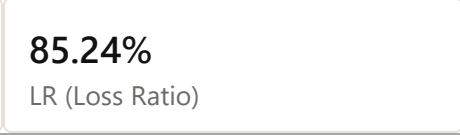
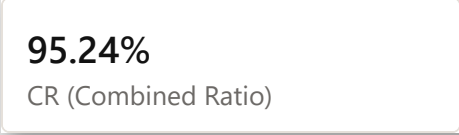
The data model extracts data from a **PostgreSQL database** and a budgeting table in Excel, using **Import mode** for dimension tables and Excel data, and **DirectQuery mode** for fact tables. The model follows a standard **star schema** design with two fact tables (Policies and Claims) and an additional budget table at a different granularity to enable **Actual vs Plan** comparisons.

The dashboard applies several effective optimisation techniques, including:

- adding **aggregation tables** (AggPolicies, AggClaims) for the fact tables,
- merging dimensions into a **star schema** from an original snowflake structure,
- removing unnecessary columns, and
- creating dedicated **date tables** and switching off auto date/time.

Usability enhancements include organising measures in a dedicated **measures table** and grouping them into **display folders**. Commission rate and expense ratio calculations are driven by **Power BI parameters**. Additional **calculated columns** are used to derive customer age (with decade-based groupings) and customer geographic location. Selected columns are grouped into **hierarchies** to support intuitive drill-down analysis, while non-essential columns are hidden to improve clarity. **Row-level security (RLS)** is implemented for lines of business (Health & Travel, Motor & Home) to ensure controlled data access. To facilitate ease of transfer to production environments, the model also incorporates **Power Query parameters** for server and database names.





3K Active Customers	1.26 Avg. LOBs per Customer	1.32 Avg. Products per Customer	964.88 Avg. GPW per Customer	5.95M Customer Lifetime Value (CLV/LTV)
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Top 7 Customers by CLV (LTV)

Customer ID	CLV (LTV)
427	6,153.77
2960	5,886.11
5409	5,864.51
3960	5,487.52
5617	5,467.17
1255	5,431.40
3522	5,387.16
Total	39,677.65

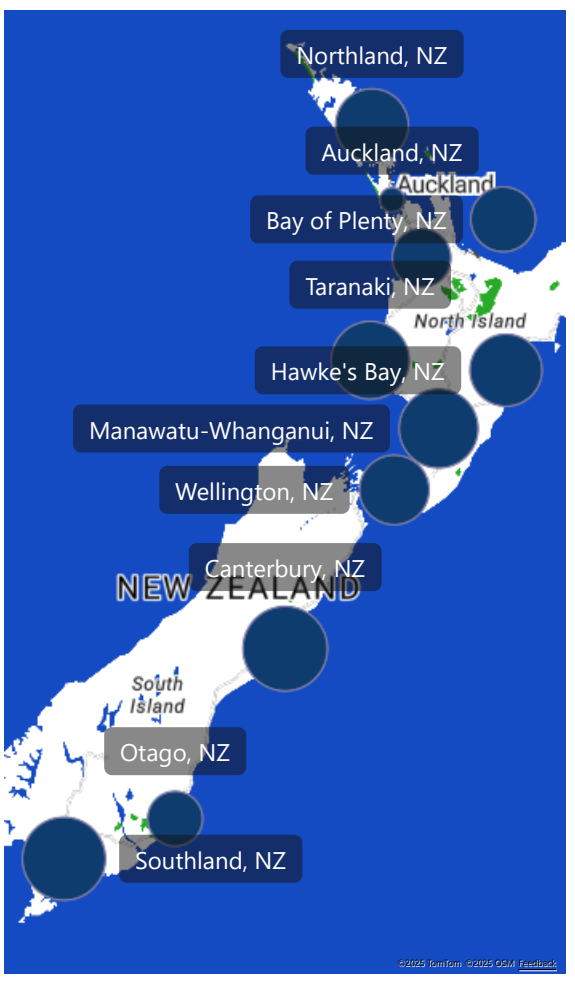
Bottom 7 Customers by CLV (LTV)

Customer ID	CLV (LTV)
786	-14,444.08
5249	-12,057.14
4940	-11,837.28
1887	-9,918.71
343	-9,519.63
533	-8,997.83
498	-8,746.90
Total	-75,521.57

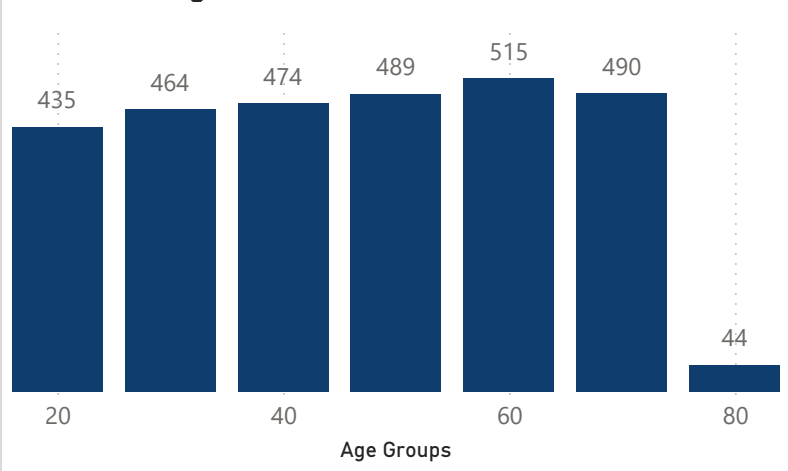
Top 7 Customers by Losses

Customer ID	Losses
5249	13,449.18
5069	10,262.51
4940	12,564.97
1887	10,618.58
786	14,742.41
498	10,103.09
343	10,626.71
Total	82,367.45

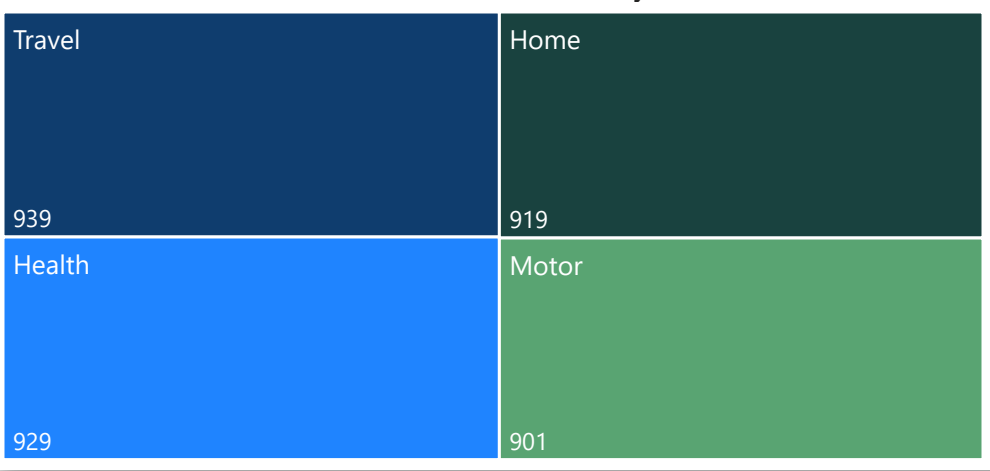
Active Customers and Premium in Force by Region



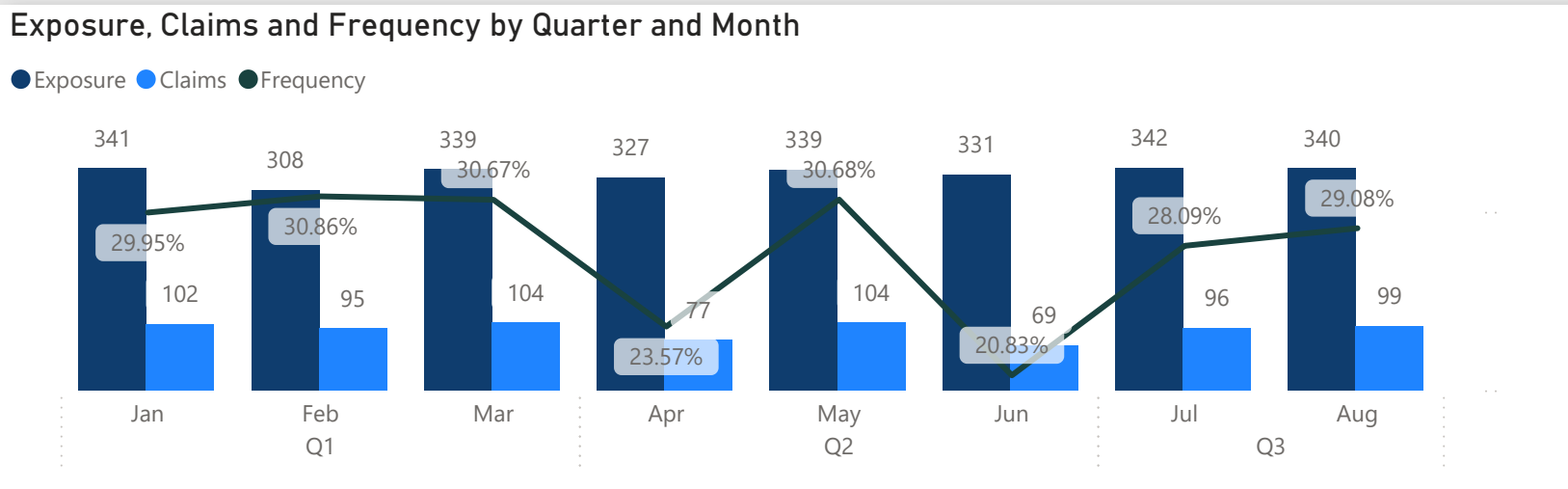
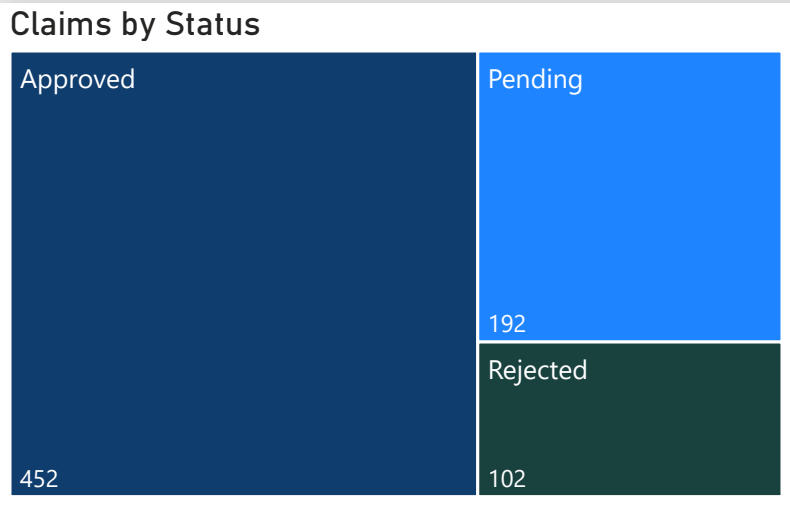
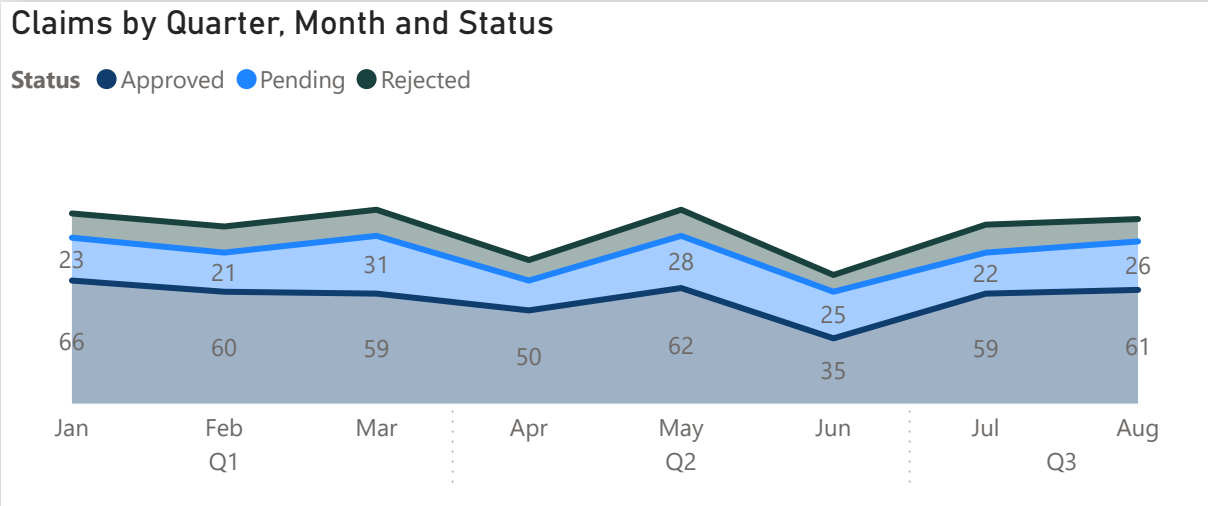
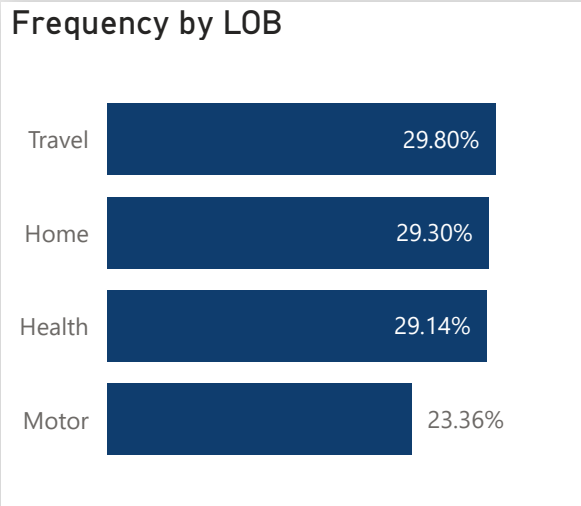
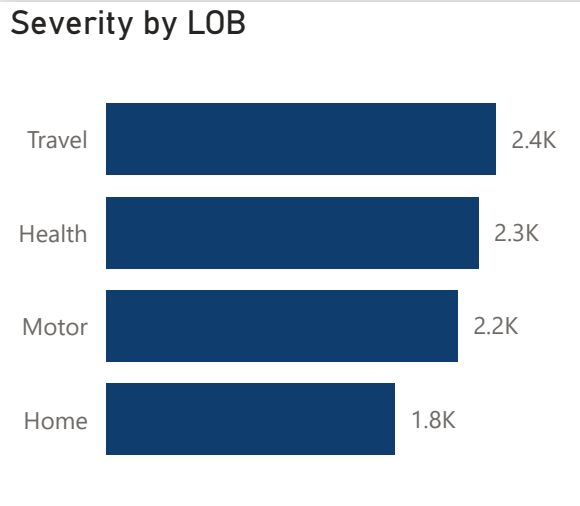
Customer Age Distribution



Active Customers and Premium in Force by LOB



477.93K	1.15M	746	27.98%	2.18K
RBNS (Reported But Not Settled)	Paid Losses	Claims	Frequency	Severity (Avg Claim Amount)



Insurance Analytics
Portfolio Project

LOB & PRODUCT PERFORMANCE

Line of Business (LOB) and Product Analysis: Gross Premium Earned (GPE), Loss Ratio (LR), Combined Ratio (CR)

95.24%

CR (Combined Ratio)

8

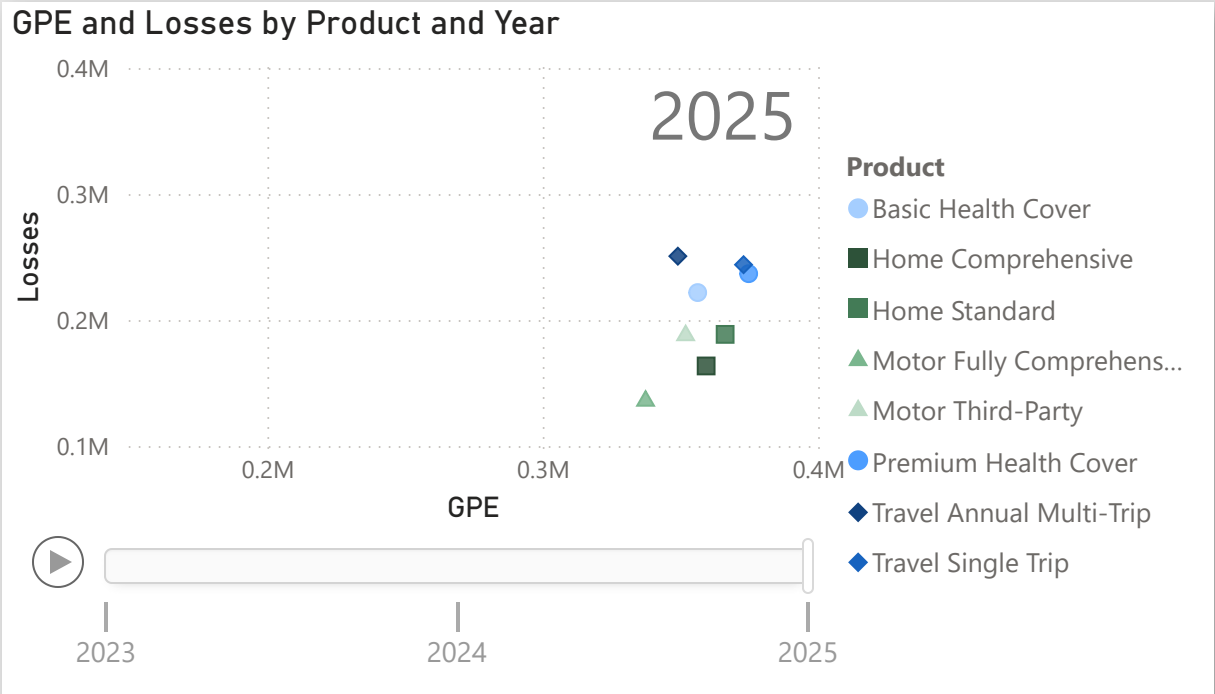
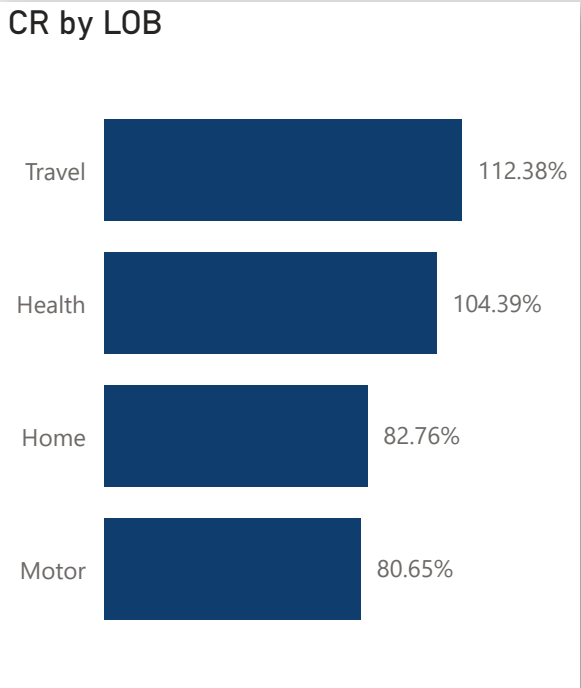
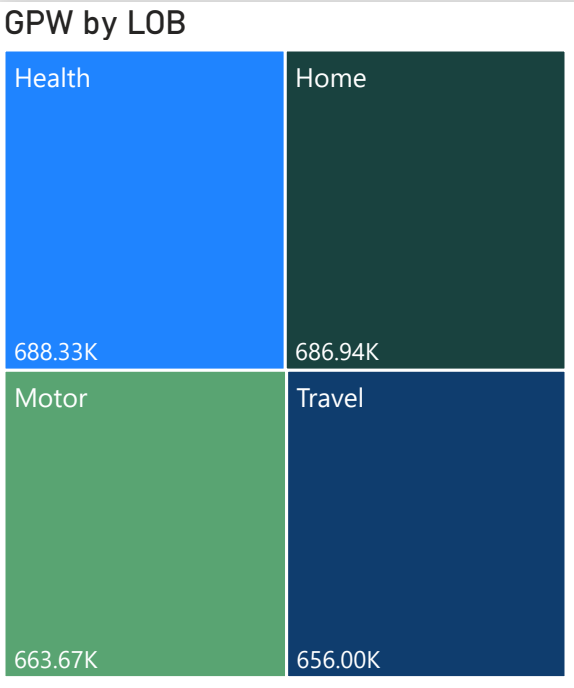
Products

4K

Active Policies

2.69M

GPW



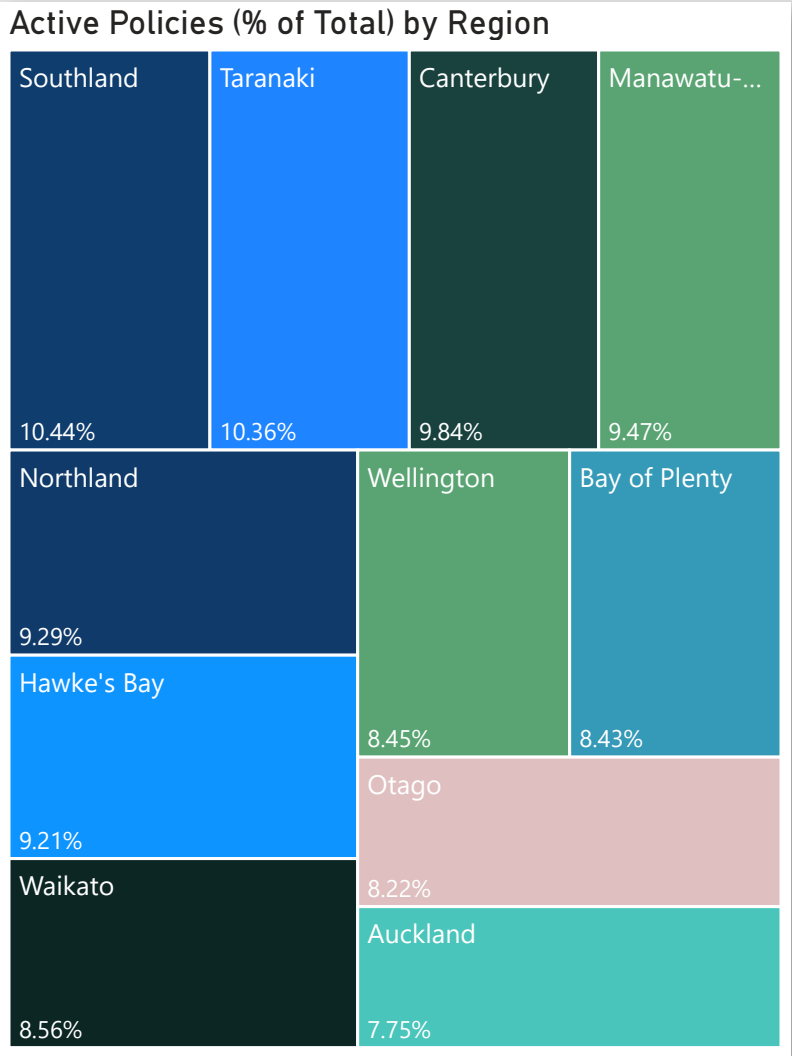
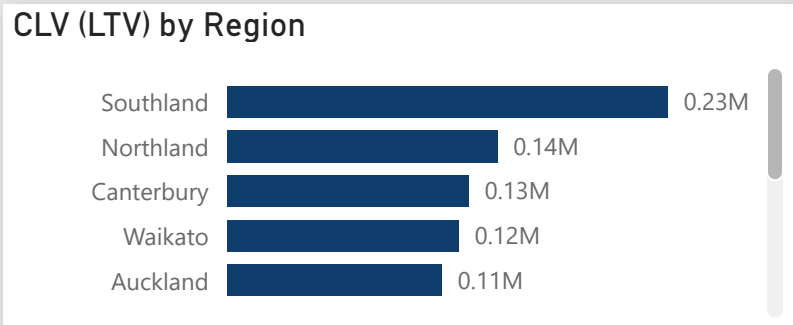
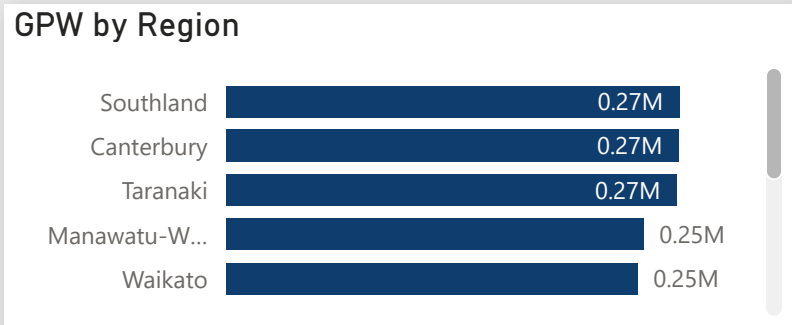
LOB	CR	Expense Ratio	Commission Rate	LR	GPE	Losses	Paid Losses	RBNS	Frequency	Severity	Claims
Health	104.39%	5.00%	5.00%	94.39%	485,776.45	458,528.60	292,697.53	165,831.07	29.14%	2,315.80	198
Home	82.76%	5.00%	5.00%	72.76%	483,951.84	352,108.44	259,557.01	92,551.43	29.30%	1,796.47	196
Motor	80.65%	5.00%	5.00%	70.65%	458,058.95	323,629.04	262,307.19	61,321.85	23.36%	2,186.68	148
Travel	112.38%	5.00%	5.00%	102.38%	483,007.86	494,526.93	336,302.01	158,224.92	29.80%	2,424.15	204
Total	95.24%	5.00%	5.00%	85.24%	1,910,795.10	1,628,793.01	1,150,863.74	477,929.27	27.98%	2,183.37	746

Southland
Top Region by GPW

273.44K
Top Region GPW

Southland
Top Region by CLV (LTV)

228.71K
Top Region CLV (LTV)



GPW by Region, LOB and Product

Region	Health	Home	Motor	Travel
Auckland	51.7K	42.8K	65.7K	42.6K
Otago	54.2K	60.0K	54.4K	56.1K
Wellington	61.5K	53.9K	53.0K	60.6K
Bay of Plenty	51.7K	61.9K	60.9K	57.7K
Hawke's Bay	66.6K	59.3K	63.3K	52.8K
Northland	62.1K	63.3K	53.1K	67.6K
Waikato	54.7K	68.6K	64.9K	59.9K
Manawatu-Whanganui	61.6K	70.0K	57.5K	62.7K
Taranaki	76.4K	62.7K	59.4K	73.2K
Canterbury	72.3K	69.3K	66.5K	64.9K
Southland	75.4K	75.1K	64.9K	58.1K

CR by Region, LOB and Product

region_name	Health	Home	Motor	Travel
Auckland	114.7%	117.5%	93.3%	81.4%
Bay of Plenty	154.9%	85.9%	52.4%	34.9%
Canterbury	71.3%	94.4%	67.4%	104.3%
Hawke's Bay	98.6%	68.5%	87.8%	114.2%
Manawatu-Whanganui	129.9%	72.3%	111.8%	168.3%
Northland	74.4%	42.0%	49.4%	133.9%
Otago	120.2%	93.5%	34.9%	130.1%
Southland	37.2%	98.3%	57.5%	60.0%
Taranaki	113.0%	85.8%	183.0%	136.0%
Waikato	56.8%	61.1%	81.5%	186.0%
Wellington	157.1%	104.5%	41.7%	77.0%